

LEBRE v0.53 — Specification and Technical Report

Consolidation of v0.52: precision expert and starting point at the trivial reference, under a declared budget

LEBRE: Lifecycle-governed Evidence-Based Resource Evolution

v0.53: unchanged v0.52 + M1 (precision expert) + M2 (trivial reference)

STATUS

research version promoted by a pre-registered rule (2026-10-09); v0.52 remains frozen and unchanged

FINAL EVALUATION

104 reserved series; $v0.53 \div v0.52 = 0.955$ [95% CI 0.951; 0.960]; no harm per family

COST

median increment of 476 operations per step over v0.52 in the final evaluation (limit 1,100); estimated memory 83.1 KB

SCOPE

gain in basins and FX; identical to v0.52 in solar, wind and load; one step ahead

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1. Executive summary

LEBRE v0.53 is the **unchanged** v0.52 with two new components on top. **M1** combines the v0.52 forecast with a precision expert (a recursive regression on the target's past and on lag-band means of the inputs), for the cases where v0.52 lies below the accuracy × cost frontier. **M2** makes the forecast start from a declared trivial reference (zero, persistence or seasonal naive) and trust the adaptive part only when it proves better, for the cases where v0.52 loses to that reference.

Result of the pre-registered final evaluation (104 series reserved before any v0.53 code):

- **promoted** by the rule fixed beforehand: no harm in any family and a clear overall improvement;
- mean squared error v0.53 ÷ v0.52 of **0.955** [IC 95% 0.951; 0.960] over the 104 series;
- the gain comes from Basins (CAMELS-BR) 0.879, FX: levels 0.952, FX: returns 0.966, Buildings (BDG2) 0.993; in solar, wind and load v0.53 is identical to v0.52;
- cost: median increment of 476 operations per step over v0.52 (maximum 918; pre-registered limit 1,100); estimated memory 83.1 KB.

Premise: as in v0.52, LEBRE does not aim to be the most accurate forecaster; it aims to deliver well under a minimal budget and without severe failures. v0.53 costs more than v0.52 on every series and only gains in part of the domains; that is the main limit of this version (§11).

2. Starting point: v0.52 and two failure modes

v0.52 (specification, revision 1) is an online forecaster with a simple live model (memory of the target itself, regression on the inputs and dynamic averaging of the two) and a plane of structural changes tested by e-processes. It was frozen without promotion on 2026-09-27. The LEBRE Lab, a public test bench, surfaced failures of v0.52 in new domains; v0.53 addresses two of them without widening the hypothesis space (the others were left for v0.54).

F8: losing to the trivial reference

In nearly unpredictable series (FX returns and levels), the adaptive part of v0.52 errs more than predicting zero or repeating the last value. In the final reserve itself, the reference beat v0.52, with the whole confidence interval above 1, in 5 of 8 return series and 6 of 8 level series.

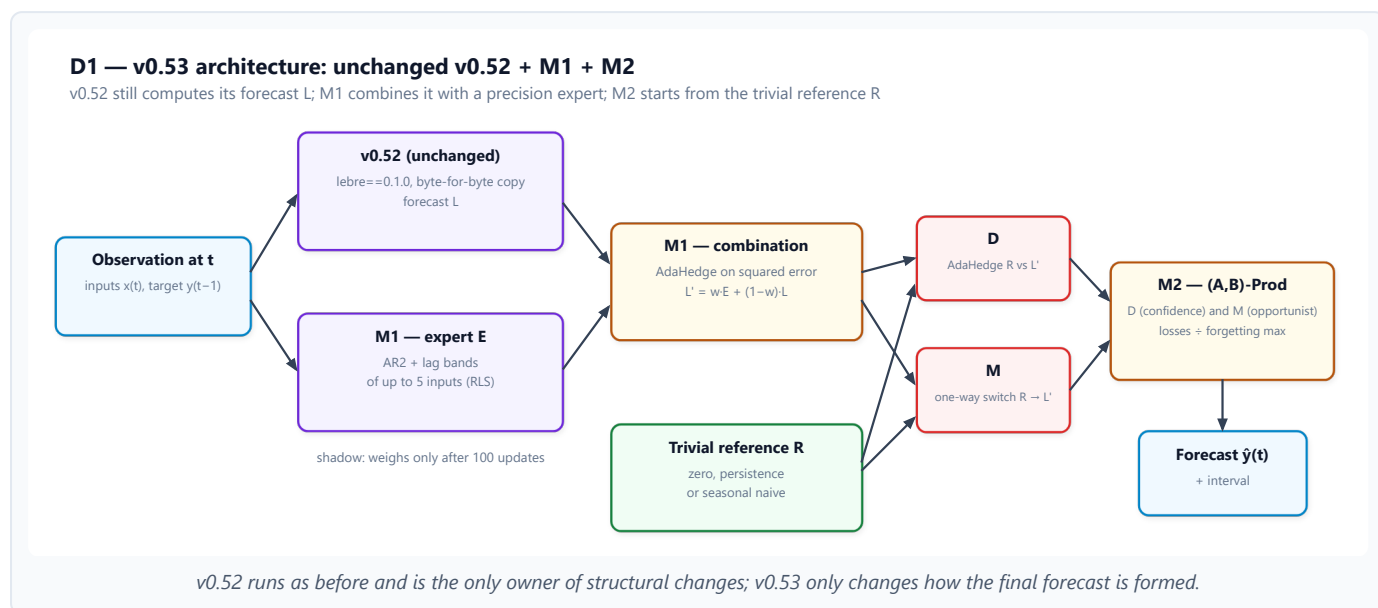
F6: below the accuracy × cost frontier

In 8 development families a full linear comparator (all inputs and lags estimated jointly, from a few thousand to tens of thousands of operations per step) beats v0.52 by more than 5%. What matters under a budget is whether v0.52 loses to something of *the same cost*. The diagnostic measured 8 regressions of varied cost in each family: v0.52 is below the frontier (a regression of equal or lower cost errs 5% less) in 4 of them, all with the same cause, the lack of joint estimation of the target dynamics and of the input effect.

Family	v0.52 ÷ full	best with cost ≤ v0.52	v0.52 cost (FP)	status
B03	2.076	1.574	731	below
VB07	1.170	1.005	435	below
XB07	1.141	1.010	417	below
B07	1.067	1.004	431	below
YB06	1.095	1.048	640	on the frontier
YB03	1.225	1.397	830	on the frontier
VC05	1.138	1.381	262	on the frontier
A05	1.077	1.777	385	on the frontier

Error = MSE ÷ MSE of the full online linear comparator (family geometric mean). B03: air quality; B07, VB07, XB07: storm surge (three independent sets); VC05: inflation; YB03: Beijing air quality; YB06: wind; A05: synthetic.

3. Architecture overview



- **Nothing in v0.52 changes.** The four base files are byte-for-byte copies of `lebre==0.1.0`; the structure v0.52 accepts inside v0.53 is the same as when it runs alone (verified on every evaluated series).
- **M1** produces L' from L and the expert E . Where E does not help, its weight goes to zero and $L' = L$.
- **M2** mixes R and L' through two paths (D , conservative; M , which switches from R to L' once) and combines both with an (A,B) -Prod, which guarantees not falling far behind D .
- All new cost is fixed per step, given the number of inputs; nothing grows with time.

4. M1: precision expert

The expert

Variables: the target itself at lags 1 and 2 and, for each of the $m = \min(d, 5)$ inputs with the largest absolute correlation with the target, the standardised means over the lag bands [0], [1], [2–3], [4–7] and [8–15]:

$$z_t = [1, \tilde{y}_{t-1}, \tilde{y}_{t-2}, \bar{x}_{i,b}(t)]_{i \leq m, b \leq 5}, \quad k = 3 + 5m, \quad E_t = w_t^\top z_t$$

Weights come from recursive least squares with forgetting $\lambda = 0.999$ and $P_0 = 100 \cdot I$, updated every 8 observed targets. Input selection uses exponential means (past only), is made after 100 targets and redone every 5,000; if the set changes, the regression and the combination restart. The forecast is clipped to $L \pm 2 \cdot \max(\hat{\sigma}_L, \text{floor})$, $\hat{\sigma}_L$ being the recent v0.52 error, which bounds the damage of an ill-conditioned regression.

The combination with v0.52

AdaHedge (de Rooij et al., 2014) with two experts, v0.52 and E, on the **squared error without clipping or normalisation**: the weights follow the accumulated squared error, which is what the MSE measures, and the learning rate adapts to the scale by itself. Shadow entry: until 100 updates, $L' = L$.

$$L'_t = (1 - \omega_t) L_t + \omega_t E_t, \quad \omega_t = \frac{e^{-\eta_t S_{t-1}^E}}{e^{-\eta_t S_{t-1}^L} + e^{-\eta_t S_{t-1}^E}}, \quad S_{t-1}^\bullet = \sum_{\tau < t} (y_\tau - \bullet_\tau)^2$$

Why not an (A,B)-Prod, as in M2: Prod needs losses in [0, 1]. Three ways of putting the squared error into that interval were measured and failed for different reasons (§8): clipping at 2σ leaves Prod blind to large errors; dividing by the largest error ever seen freezes it after a record; dividing by the largest recent error gives small weight to the large-error steps, exactly where the expert gains. Trading Prod's constant guarantee for AdaHedge's (a regret that, relative to the total MSE, vanishes with series length) is what worked.

5. M2: starting from the trivial reference

The reference R is declared with the data: zero for financial returns, seasonal naive when there is a cycle, persistence otherwise. While R is undefined (before a full cycle), the output is L' .

- **D** — AdaHedge between R and L' on the squared error (only on steps with R defined): confidence, no switches.
- **M** — one-way switch (switch distribution, van Erven et al., 2012): posterior over “always R ”, “always L' ” and “ R until some time, L' after”, with switch hazard $1/t$, on Gaussian losses at the recent scale; a log-scale path avoids 0/0 when all the mass is on one side.
- **Combination** — anytime (A,B)-Prod (Sani et al., 2014) with $B = D$ (confidence) and $A = M$ (opportunistic); losses $(y - \cdot)^2/N(t)$, with $N(t) = \max(\text{step losses}, 0.99 \cdot N(t-1))$.

$$\hat{y}_t = s_t M_t + (1 - s_t) D_t, \quad s_t = \frac{\eta_t w_t^A}{\eta_t w_t^A + w^B/2}, \quad \eta_t = \min\left(\frac{1}{2}, \sqrt{1/(1 + \sum_{\tau < t} \delta_\tau^2)}\right)$$

Guarantees and the limit of shocks

For the normalised losses, Prod guarantees constant regret against D and of order $\sqrt{(T \log \log T)}$ against M ; in the original scale there is no clean formal bound, because the normalisation changes over time. There is a limit no online combiner escapes: guarantees for unbounded losses hold in units of the largest single-step loss, and in financial series one day can account for much of the whole error (in the 521 measured series, 32 have the largest step above 5% of the reference's total error; the largest, baht_1990, 25.9%). That is why the evaluation ruler treats those series as undecidable (§7).

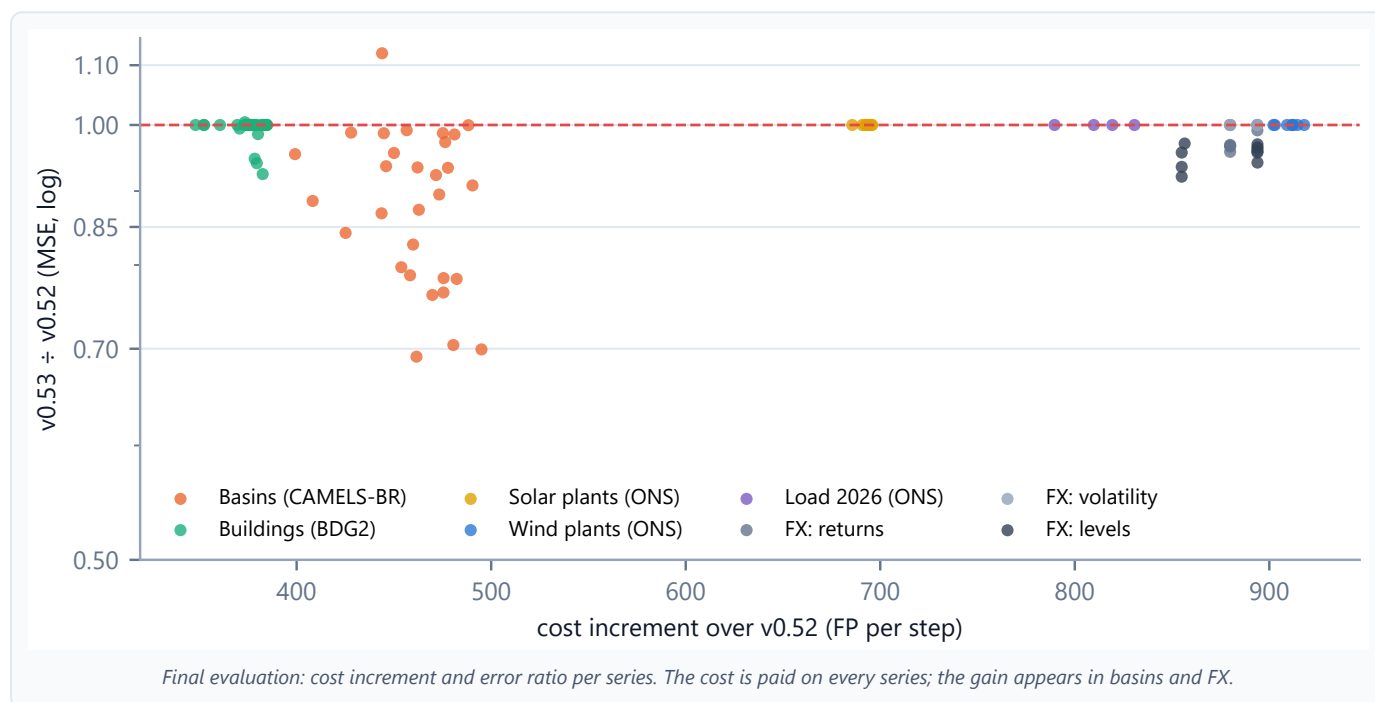
6. Computational cost

Count by the v0.52 rule (one elementary operation = 1 FP), audited: the implementation performs exactly the declared operations (each pair of the RLS matrix symmetrised once; only the new input row standardised), with forecasts identical to the validated ones in 813 of 813 series; the audit raised the count by 3.2% (median).

Part	FP per step (median, one series per family, before the audit)
v0.52	453
M2	137
M1: RLS update	513
M1: rest	146
Total	1,257

RLS: $6k^2 + 5k$ per update (every 8 targets); expert prediction: $2k + 18m + 4$; AdaHedge: 48; M2: ~132 to 137. In development and validation 5 (813 series), the increment over v0.52 had a median of 538 and a maximum of 974 FP; the ratio $v0.53 \div v0.52$, a median of 2.32 and a maximum of 3.93.

The preliminary target of the design note (cost ≤ 1.25 times that of v0.52) is incompatible with M1 + M2: with v0.52 at ~450 FP, the maximum increment would be ~110 FP, less than M2 alone. Before the final evaluation, the project owner replaced it with a ceiling of an **absolute increment of 1,100 FP per step** on every series (the new cost is nearly fixed; a ratio penalises the series where v0.52 is cheap), plus 2.5 times on F6-type families.



7. Evaluation methodology

Discipline

- The final-evaluation data were reserved on 2026-10-05, before any v0.53 code, with hashes and locks in both repositories.
- Every measurement has its plan and criteria committed beforehand; nothing is adjusted afterwards; failures stay declared.
- Each validation set is chosen from metadata only, frozen with SHA-256 and used once.

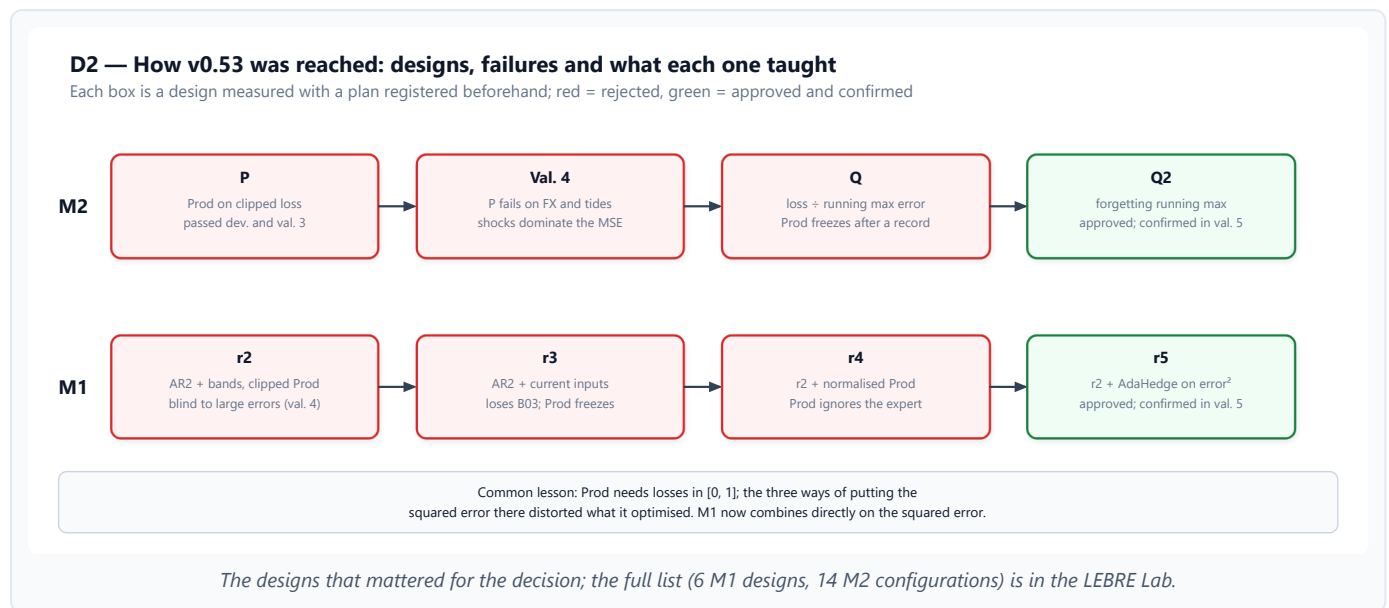
Decidability ruler

Each series is classified by the 95% CI of $v0.52 \div R$ (paired block bootstrap, 2,000 replicates): “v0.52 better”, “reference better” or “undecidable”. Criteria: (1) on “reference better”, $v0.53 \div R$ with upper limit ≤ 1.01 ; (2) on “v0.52 better”, $v0.53 \div v0.52$ with upper limit ≤ 1.02 ; (4) on negative tests, no more false inputs than v0.52; (6) v0.52’s structure preserved; (7) on short series (360 steps), ≤ 1.05 of the best of v0.52 and R. Three addenda, decided during development and before validation 5: a series whose largest single step exceeds $\max(5\%, 5 \cdot 2 \ln(n)/n)$ of R’s error is undecidable on criteria 1 and 2 (n = evaluated steps).

F6 and promotion

F6, on the families below the frontier: cost ≤ 2.5 times that of v0.52 and error ≤ 1.05 times the lowest error among the diagnostic’s 8 regressions with cost $\leq$ that of v0.53. Promotion (pre-registration, project-owner decisions): (A) no harm per family, (B) $v0.53 \div v0.52$ over all series with the upper limit of the 95% CI below 1 and (C) estimated memory ≤ 128 KB.

8. Development: designs and failures



The development bench grew to 64 families and 728 series (original development and validations 1 to 4, already spent, plus the short families). On it, the final combination (M1 draft 5 + M2 Q2) met all M2 criteria and F6 on the four families below the frontier:

Family	v0.52 ÷ full	v0.53 ÷ full	target	cost ÷ v0.52
B03	2.076	1.365	1.652	2.01
B07	1.067	1.020	1.054	2.17
VB07	1.170	1.053	1.056	2.13
XB07	1.141	1.044	1.061	2.21

Process findings, all recorded: a numerical defect in component M (0/0 by underflow) was found in a test, fixed and audited (it never fired in a recorded measurement; since then a non-finite forecast counts as a failure); a construction error in validations 4 and 5 (30 h cycle instead of 24 h in solar, wind and load) restricted the scope of validation 5 (§9).

9. Validations on new data

Validations 1 and 2 served earlier M2 designs; validation 3 confirmed candidate P. **Validation 4** rejected M1 draft 2 with P: criterion 1 on YB07, YC01 and criterion 2 on YB01. The diagnostic separated the causes (the solar failure came from M1; the FX and tide ones from M2) and led to the designs of §8.

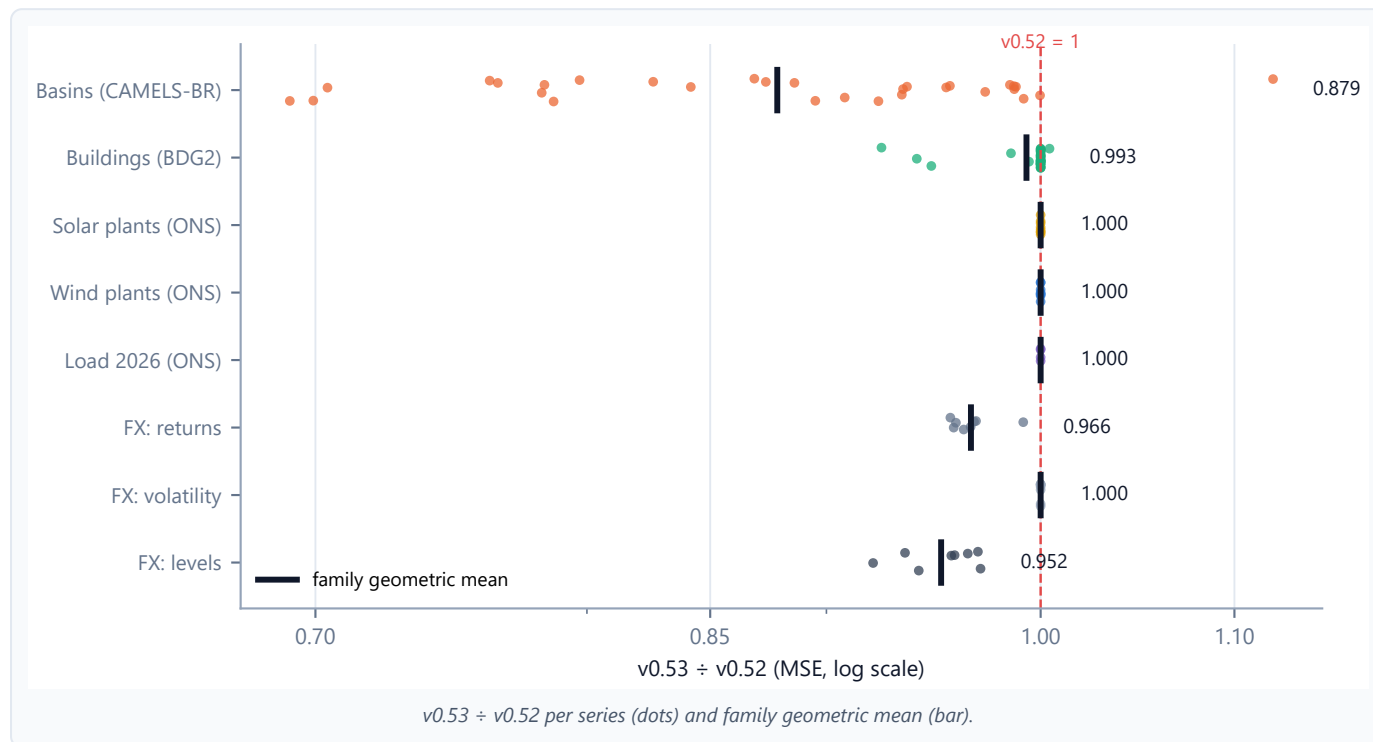
Validation 5 (43 new series, single measurement) confirmed M1 draft 5 with M2 Q2: F6 met on Beijing air quality (Chen, 2017) and on six new tide stations, and all M2 criteria on every family.

Family	v0.52 ÷ full	v0.53 ÷ full	target	cost ÷ v0.52
Z-B03	1.169	1.118	1.410	1.97
Z-B07	1.017	0.925	1.068	2.20

Errata: in validations 4 and 5, the solar, wind and load families were built with a 30 h cycle instead of 24 h. The v0.52 vs v0.53 comparisons in those families hold for the task as built, not for the declared analogue; the fresh evidence for v0.53 with the right cycle in those domains is the final evaluation (§10).

10. Pre-registered final evaluation

Single run on 104 reserved series (30 CAMELS-BR basins, Chagas et al., 2020; 30 BDG2 meters, Miller et al., 2020; 8 solar plants, 8 wind plants, the 2026 load of the 4 subsystems and 8 FX pairs in 3 tasks). The pre-registration was approved after a rehearsal on 55 already spent basin and building series (0 errors).



Family	series	v0.52 better / ref. / undec. / shock	$v0.53 \div v0.52$	crit. 1	crit. 2	max +cost (FP)
Basins (CAMELS-BR)	30	15/2/7/6	0.879	0.845 [0.814; 0.878]	0.911 [0.899; 0.924]	495
Buildings (BDG2)	30	30/0/0/0	0.993	-	0.993 [0.988; 1.000]	385
Solar plants (ONS)	8	8/0/0/0	1.000	-	1.000 [1.000; 1.000]	696
Wind plants (ONS)	8	8/0/0/0	1.000	-	1.000 [1.000; 1.000]	918
Load 2026 (ONS)	4	4/0/0/0	1.000	-	1.000 [1.000; 1.000]	831
FX: returns	8	0/5/1/2	0.966	1.000 [1.000; 1.000]	-	894
FX: volatility	8	6/0/0/2	1.000	-	1.000 [1.000; 1.000]	894
FX: levels	8	0/6/0/2	0.952	1.001 [1.000; 1.002]	-	894

- Overall improvement: **0.955 [0.951; 0.960]**; short series against the best of v0.52 and R: 1.007 [0.995; 1.017].
- No non-finite forecast; false inputs on returns equal to v0.52's; structure preserved on every series; estimated memory 85,092 B.
- In solar, wind and load, M1 ended with zero weight on all 20 series and v0.53 was identical to v0.52.
- Two fixes during the run, with no effect on model or criterion: reading the 2026 load by subsystem code (ONS renamed the Southeast) and handling a degenerate short series (exact tie with zero error).

Reference ceilings (reported only)

Models outside the budget class, as reference only: Chronos-2 with covariates (Ansari et al., 2025), a foundation model of $\sim 10^9$ operations per forecast, on the 1,000 points of the v0.52 protocol; and SARIMAX with inputs.

Family	v0.53 ÷ Chronos-2 (1,000 points)	v0.52 ÷ Chronos-2	SARIMAX-X ÷ v0.53
Basins (CAMELS-BR)	0.885	1.008	1.134
Buildings (BDG2)	1.248	1.241	1.034
Solar plants (ONS)	1.237	1.237	1.096
Wind plants (ONS)	1.051	1.051	1.049
Load 2026 (ONS)	1.433	1.433	0.968
FX: returns	0.954	0.979	1.123*
FX: volatility	0.983	0.983	1.058*
FX: levels	0.881	0.921	1.065*

* FX: SARIMAX with inputs did not run in the execution (inputs with missing holidays); values from the addendum with inputs filled by the v0.52 rule, made after the result.

11. Limits and claims we do not make

- **Cost without gain in part of the domains:** v0.53 costs more on every series; in solar, wind and load it is identical to v0.52.
- **F6 without a family in the final reserve:** the evidence for F6 on new data is validation 5 only; one development family passed by a small margin (VB07).
- **Ruler decided during development:** the shock addenda were decided after seeing development data, before validation 5 and the final evaluation.
- **No formal guarantee in the original scale:** Prod guarantees on normalised losses; M1's AdaHedge gives no constant regret against v0.52.
- **Memory estimated, not measured;** no C port nor microcontroller measurement.
- **Uncertainty:** the overall CI treats series as independent; load has 4 series.
- **Errata** in validations 4 and 5 and two fixes in the final run, all declared.
- **Originality:** no component is new; the contribution is the integration under a declared budget and the pre-registered evidence.
- **Outside this version:** removals that happen, cost proportional to use and cycle as hypothesis (v0.54).

12. Position in the literature

Every part has a direct antecedent: AdaHedge and FlipFlop (de Rooij et al., 2014); (A,B)-Prod and constant regret against a trusted expert (Sani et al., 2014; Even-Dar et al., 2008; Koolen, 2013); switch distribution (van Erven et al., 2012) and Fixed Share (Adamskiy et al., 2016; Cesa-Bianchi et al., 2012); recursive regression with lags (Ljung, 1999). Guarantee limits for unbounded and heavy-tailed losses are in Cesa-Bianchi et al. (2007), Hazan and Kale (2010), Orabona and Pál (2018), Mhammedi et al. (2019), V'yugin and Trunov (2019), Alquier (2021) and Moulin et al. (2025); heavy-tail-robust confidence sequences, in Wang and Ramdas (2023), remained a direction for M2. The contribution of this version is the **integration** of these components under a declared budget and the pre-registered evaluation.

13. Parameters

Part	Parameter	Value	Note
M1	inputs	$m = \min(d, 5)$	largest absolute correlation (exponential means, past only)
M1	bands	[0], [1], [2–3], [4–7], [8–15]	standardised means
M1	RLS	$\lambda = 0.999$; $\delta = 100$	every 8 observed targets
M1	selection	after 100; every 5,000	restarts regression and combination
M1	clip	$L \pm 2 \cdot \max(\hat{\sigma}_L^2, \text{floor})$	$\hat{\sigma}_L^2$: recent v0.52 error
M1	combination	AdaHedge	squared error; 100-update shadow
M2	R	zero / seasonal / persistence	declared with the data
M2	D	AdaHedge	squared error, only with R defined
M2	M	one-way switch, hazard $1/t$	Gaussian losses at the recent scale
M2	Prod	$\eta \leq 1/2$; $w_B = 1/2$	losses $\div \max(\text{losses}, 0.99 \cdot N)$
M2	gate	$\varepsilon = 0.002$; $\alpha = 0.01$	decision audit; recreated after each decision

The parameters of the v0.52 base are those of the v0.52 specification, revision 1, §20. Nothing is tuned per series.

14. Pseudocode and reproducibility

```

CONSTANTS  M_MAX = 5   BANDS = [0], [1], [2-3], [4-7], [8-15]    $\lambda_{\text{RLS}} = 0.999$     $\delta_{\text{RLS}} = 100$    EVERY = 8
              CLIP_K = 2   N_MIN = 100   T_MAX = 5000   LAM = 0.99    $\epsilon_{\text{gate}} = 0.002$     $\alpha_{\text{gate}} = 0.01$ 

STEP( $x_t$ ,  $y_{t-1}$ ):
  L  $\leftarrow$  v052.predict( $x_t$ )                                # the frozen base, unchanged
  # M1: precision expert
  z  $\leftarrow$  [1,  $\hat{y}_{t-1}$ ,  $\hat{y}_{t-2}$ , standardised BAND means of the m = min(d, 5) chosen inputs]
  E  $\leftarrow$  clip( $w \cdot z$ , L  $\pm$  CLIP_K  $\cdot$  max( $\sigma_t^2$ , floor))    if RLS determined (k updates)
  L'  $\leftarrow$  L if AdaHedge.n < N_MIN else  $w_L \cdot L + w_E \cdot E$     # shadow entry
  # M2: start from the trivial reference
  R  $\leftarrow$  0 |  $y_{\text{last}}$  |  $y_{t-s}$                                 # zero, persistence or seasonal naive
  D  $\leftarrow$   $w_D \cdot [R, L']$ ;   M  $\leftarrow$   $w_M \cdot [R, L']$ ;   s  $\leftarrow$   $\eta \cdot w_A / (\eta \cdot w_A + w_B / 2)$ 
   $\hat{y} \leftarrow$  L' if R undefined else s  $\cdot$  M + (1 - s)  $\cdot$  D
  return  $\hat{y}$ 

OBSERVE( $y_t$ ):
  v052.observe( $y_t$ ); screening every EVERY; RLS every EVERY targets; new selection after N_MIN and every T_MAX
  AdaHedge.update(( $y_t - L$ )2, ( $y_t - E$ )2)                # squared error, no clipping
  AdaHedge_D.update(( $y_t - R$ )2, ( $y_t - L'$ )2)
  Switch_M.update(( $y_t - R$ )2/2 $\sigma^2$ , ( $y_t - L'$ )2/2 $\sigma^2$ )    # recent  $\sigma^2$ ; log-scale fallback against 0/0
  N  $\leftarrow$  max(( $y_t - D$ )2, ( $y_t - M$ )2, LAM  $\cdot$  N); Prod.update(( $y_t - D$ )2/N, ( $y_t - M$ )2/N)

```

- **Promoted code:** research repository (lebre-research), commit 4a2620e, v0.53 prototype folder; canonical configuration in the v0.53 freeze record.
- **Integrity:** the v0.53 SHA-256 manifest covers 177 files; the repository integrity verifier checks every frozen version.
- **Final evaluation:** the v0.53 final-evaluation folder of the research repository (pre-registration, scripts, run notes, results).
- **Development and validations:** LEBRE Lab, github.com/Iqitim/lebre-lab (plans before each measurement, results, diagnostics, errata); this document read the Lab results at commit 6f328bb.
- **Environment:** Python 3.11, numpy 2.2.5, pandas 2.2.3, statsmodels 0.15.0; Chronos-2 with the chronos 2.3.2 package.
- **Determinism:** no random numbers in the model; the same input sequence gives the same output.

15. References

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16. Revision history

- **Revision 0** (2026-10-09): first edition, after the promotion.